3.1 - Extreme Values

Goals: Identify and find absolute and relative extrema, particularly on closed intervals.
Before Class Videos: 3.1.1, 3.1.2

Definition:

Let f be defined on an interval I containing c.

1. $f(x)$ is the **absolute** (or global) maximum if $f(c) \geq f(x)$ for all x in I.
2. $f(x)$ is the **absolute** (or global) minimum if $f(c) \leq f(x)$ for all x in I.
3. $f(x)$ is a **relative** (or local) maximum if $f(c) > f(x)$ for all x in some interval about c within I.
4. $f(x)$ is a **relative** (or local) minimum if $f(c) < f(x)$ for all x in some interval about c within I.

Theorem (Extreme Value Theorem):

On a closed interval $[a, b]$, a continuous function must attain an absolute maximum and absolute minimum in the interval.

Example 1: Sketch examples where the assumption is not satisfied and the conclusion of the Extreme Value Theorem is false.

Example 2: What are the maximum and minimum values of $f(x) = x^2 - 6x + 7$ on $[2, 5]$.

Definition:
A **critical number** of f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist. A **critical point** is a pair $(c, f(c))$ when c is a critical number.

Example 3: Find the absolute maximum and minimum of $f(x) = x^{2/3}$ on $[-1, 1]$.

Extreme Value Method for Closed Intervals:

1. Find all critical points: $f'(x) = 0$ or DNE.
2. Evaluate $f(x)$ at each critical point (in the interval) and endpoints of the interval
3. Pick biggest/smallest.

Example 4: Find the absolute max and min of $f(x) = x^4 - 8x^2$ on $[-1, 3]$.

Example 5: $f(x) = x^{1/3}$ on $[-1, 1]$.

Theorem:
If f has a max at c and $f'(c)$ exists, then $f'(c) = 0$.

Example 6: $f(x) = \ln(x+1) + \tan x$ on $(-1, 1]$.

Suggested after class problems from APEX: Section 3.1: 7,10,17,19